

Soil bulk density from organic carbon and sampled depth

The source records soil health measurements at eight European agroforestry sites. The prediction endpoint is measured dry-soil bulk density; carbon stocks are excluded because their calculation contains that endpoint. The frozen complete-case contract requires organic carbon, interval depth and water pH. It retains 430 rows from six countries: four for development and Czechia/Germany for confirmation. UK and the Netherlands have no water-pH measurements in this workbook and are excluded transparently.

The development-selected two-phase volume relation fails to transfer reliably. Its reserved MAE is 0.336 g/cm³, compared with 0.173 for the simple mean-density control. This portfolio therefore contributes a falsification and a bounded evaluation task, not a positive pedotransfer law. No candidate was promoted after confirmation.

For carbon mass percentage *C*, the conventional approximate organic fraction is $f = 1.724C/100$. The volume-mixture equation below uses effective component bulk densities, not measured intrinsic grain densities. Organic conversion and constant packing are modeling assumptions. Empty German and Czech country-metadata cells prevent stronger attribution of the transfer failure; native carbon units remain exactly the published column labels.

Experimental scope and evaluation

Same soil-sampling occasion; Corg, sampling depth and pH are measured covariates available before BD inference. This is a laboratory-measurement substitution task, not a future forecast.

Calibration: No held-out country BD calibration; country identifiers are grouping metadata, never predictors.

Validation unit: Entire country/site portfolio, with all transects, fields and depths linked. Eight countries are convenience samples, not a random population sample.

Country-balanced errors and explicit individual-country results; two confirmation countries are insufficient for precise population confidence.

Target: measured soil bulk density (g/cm³). Errors use g/cm³. The selected reference is **two_phase_volume**. Selection was frozen before confirmation; diagnostic winners are not substituted afterward.

Predictor	Unit
C	mass % organic carbon
depth	cm midpoint of sampled interval
pH	pH unit in water

Matched comparison

Model	Development MAE	Reserved MAE	Reserved worst-group MAE
constant	0.369453	0.172601	0.21367
flexible	0.26609	0.314545	0.556889
linear	0.319091	0.312858	0.573624
two_phase_volume	0.259694	0.336125	0.608326
structural_porosity	0.326218	0.337788	0.615075
depth_compaction	0.312393	0.323582	0.578477
organic_consolidation	0.297844	0.326608	0.594189
acidity_aggregation	0.326262	0.359424	0.621493

The primary error is mean group MAE. RMSE and signed bias are complementary; rows within a group do not establish independent replication. No accuracy percentage or industrial tolerance is invented.

Selected equation and coefficients

$$\hat{\rho} = \left[\frac{1-f}{\rho_m} + \frac{f}{\rho_o} \right]^{-1}, \quad f = 1.724C/100$$

C is the source carbon percentage used by the declared surrogate, with the conventional 1.724 conversion. The source ambiguity and failed country transfer preclude interpreting this as a validated density law. Density parameters use g/cm^3.

Coefficient	Frozen value
mineral	1.59923279
organic	0.142370917

All comparator expressions, numerical guards and full-precision values remain in EQUATIONS.md, rules.json and run.py.

Applicability and limitations

- Texture is unavailable in six countries and excluded rather than imputed. C stocks are excluded because they use bulk density algebraically.
- Observational associations do not identify an agroforestry treatment effect. Protocol adaptations and country effects may limit transfer.

Two Finland rows were seen during schema inspection, so Finland is development-only. Reserved countries were allocated by metadata-only hash before target inspection. All outputs are exposed after this campaign.

Reproduction

Run python run.py in this final package to verify hashes and reproduce every saved prediction. Use python run.py --inputs new.csv --output predictions.csv --model reference only with the declared inputs and units. Numerical prediction evaluation does not certify novelty or mechanism. Source anchors and reserved observations are separate from predictor inputs.

Source: <https://zenodo.org/records/21204617>. See source/units audit and prior-art records in research history.